

Radiation-Induced Optic Neuropathy: Observation versus Intravitreal Treatment: Can Visual Acuity Be Maintained by Intravitreal Treatment?



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• **PURPOSE:** To compare intravitreal therapy with the natural course of radiation optic neuropathy after primary proton beam therapy for choroidal melanoma with respect to long-term visual acuity and development of optic atrophy.

• **DESIGN:** Retrospective comparative case series.

• **METHODS:** Inclusion criteria: patients treated with primary proton beam therapy for choroidal melanoma with a minimum follow-up of 24 months after the occurrence of radiation optic neuropathy and optic disc imaging during follow-up. Exclusion criteria: pathologic condition of the optic disc before irradiation and intravitreal therapy to treat cystoid macular edema not originating from the optic disc.

• **RESULTS:** Of 93 patients, 48 were observed only after radiation optic neuropathy, and 45 were treated with intravitreal therapy (triamcinolone, bevacizumab, and/or dexamethasone). Median follow-up was 55 months (29-187 months); median interval between onset of radiation optic neuropathy and the last patient visit was 34 months (24-125 months). Of 48 observed patients, 41 (85.4%) developed an optic atrophy after a median of 14 months (3-86 months) after radiation optic neuropathy; and of 45 intravitreally treated patients, 34 (75.5%) presented with an optic atrophy after a median of 12.5 months (1-55 months) following optic neuropathy, indicating no statistically significant differences between the groups. Comparing the change in visual acuity from occurrence of optic neuropathy to final visual acuity, no statistically significant differences were found between either group ($P = 0.579$).

• **CONCLUSIONS:** Patients treated with intravitreal therapy for radiation optic neuropathy showed no statistically significant differences related to visual acuity or optic atrophy development from patients who underwent only observation. (Am J Ophthalmol 2019;208:289-294. © 2019 Elsevier Inc. All rights reserved.)

PATIENTS WHO RECEIVE OPTIC DISC IRRADIATION due to the localization of their uveal melanoma are at high risk of developing radiation-induced optic neuropathy. The higher the radiation dose at the optic disc, the higher the risk.¹⁻³ Other risk factors such as age and female sex seem to play only a minor role.^{2,4}

Therapeutically, there is still no consensus as to whether intravitreal injections with steroids or antivascular endothelial growth factor will bring long-term advantages in terms of visual acuity compared with a purely observational approach. In the present authors' oncology service, the intravitreal therapy option was offered to patients undergoing radiation-induced optic neuropathy but with reference to a lack of studies. Based on the patient's decision, 2 groups emerged, which will be retrospectively compared in this study.

This study, therefore, examined patients with uveal melanoma who developed proton beam radiation-induced optic neuropathy to show whether and to what extent long-term visual acuity resulted after intravitreal therapy compared to patients who did not have intravitreal therapy.

SUBJECTS AND METHODS

THIS RETROSPECTIVE COMPARATIVE CASE SERIES WAS approved by the institutional review board of the Charité-Universitätsmedizin Berlin, Berlin, Germany, and was carried out in accordance with the tenets of the Declaration of Helsinki. Initially, all patients who exhibited radiation optic neuropathy with clinical signs of optic neuropathy (swelling, hemorrhage, surrounding exudates, or subretinal fluid) after proton beam therapy for

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